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From Data to Cup – Visualization Technique with 3-D Clay Printer

As ceramic artists, we consistently explore new concepts of form, especially when innovative technologies come into play. When Long Island University received its first and only 3–D clay printer in 2022, we were thrilled. Soon after the arrival of the printer, I was assigned to lead research and study of the device, given my extensive background in Computer Science and Software Development. The introduction of the 3–D printer into our academic environment led to several intriguing ideas, including its integration into various educational settings.

Although it excels at reproducing pre-fabricated data sourced from the internet with precision, crafting cookie-cutter forms lacks the artistic challenge I seek. Although employing 3–D modeling software is a common starting point for designing unique pieces, my proficiency in digital 3–D design remains somewhat limited.

In my day-to-day role in Information Technology, I often visualize data from various sources for business analysis. In this article, my goal is to introduce the visualization techniques I've used in my works, demonstrating how 3–D clay printers can serve as tools for creating art objects where "the data dictates the form."

Exploring Various Data to Forms

Visualizing numerical data is a standard practice in the Information Technology field. I often create such reports in my daily work as a software developer. However, many 3–D visualizations in these graphs, like 3–D pie charts, are generally used as eye-catching elements to highlight business analysis results. After graduating in Computer Science, I became interested in Geographic Information Systems (GIS). This interest led me to take another graduate course in Earth Science, where I encountered some compelling applications.

During this period, my thesis advisors explored and wrote about rendering daily temperature data as a spiral form in 3–D space.[1] Similarly, I later discovered that NASA had published a similar spiral animation[2] based on Ed Hawkins' "Climate Spiral" concept.[3]

Fueled by these experiences, I've set my sights on crafting unique 3–D art pieces based on time-series data, such as historical global temperature data, glacier mass data, and a three-year dataset of my Electric Vehicle's (EV) energy consumption.

KIICHI TAKEUCHI, *Not for Tomato Soup Cup*, 2022, Porcelain, 10cm x 10cm x 10cm

The concept behind this spiral form is to turn numerical temperature data into a tangible art object. Specifically, it represents monthly mean temperature anomalies from 1880 to the present. You may notice the line subtly climbing, reflecting a temperature increase of 1.5°C over 140 years. Each layer corresponds to a single year, and this curve is visibly manifested on the cup's wall. Think of this spiral form as similar to the rings of a tree.

The printer head moves in a circular motion, depositing the slip, layer by layer, as it corresponds to each month of a year. The diameter of the ring on each layer represents the temperature at that time. If the temperature increases, the ring expands on that particular layer. On the other hand, if the temperature drops, the ring contracts. Despite fluctuations between warmer and colder years, you'll see that the rim gradually expands as you move upward.

After the printing was complete, I attached the base and moved it for bisque firing. I carefully applied underglaze to highlight the temperature variations from bottom to top. To make it functional, I coated the piece with a glossy clear glaze. Following the glaze firing, I added a layer of gold low-fire glaze to signify our current era, requiring a second firing. One of my studio mates, who is majoring in Earth Science, offered this ironic comment: "Drinking beer from this cup, we can toast humanity's doomed future. Cheers!"

The other two pieces follow a similar process but focus on different datasets. For the artwork based on glacier data, the form narrows from bottom to top, reflecting the global decline in glacier mass between 1945 and 2014. I added decorative twists to the spiral form to make it resemble an upside-down icicle, envisioning it as a flower vase. I am planning to collaborate with an Ikebana artist in the future, ideally using a flower from the Arctic region for display.

The small sake sipper, or *guinomi*, represents my electric vehicle (EV) energy consumption over the past three years. I participated in the SmartCharge New York Program to share my EV driving data, and it encourages off-peak charging. Downloading daily statistics from this program, I was interested in how my driving patterns changed during the COVID-19 lockdown. The guinomi features a narrow grip in the middle, illustrating the period when I worked from home for six months and didn't drive long distances. This resulted in less energy consumption, making the diameter of the middle layers smaller. The project serves as a memorable representation of the anxiety experienced during that period, and like many others, my alcohol consumption may have also increased due to economic concerns.

Not limited to environmental data, I began exploring datasets from diverse fields. My curiosity led me to health-related metrics like my daily step count, as well as to the frequency distribution in music data. For example, what distinguishes Thelonious Monk apart from Chopin? When it comes to employing acoustic and vocal data in 3–D printed ceramic art, I'd like to highlight two previous artworks that offer contrasting approaches and yield different outcomes. Chris Gustin crafted a vessel based on the vocal data of a singer during his residency at Medalta.[4] According to his website, the form of the tea bowl directly corresponds to the audio in a 1:1 ratio. Gustin aimed for an organic form rather than a modeled shape. In contrast, Timea Tihanyi, in collaboration with Audrey Desjardins, conducted a research project to capture human activity data, including environmental sounds, using smart home

KIICHI TAKEUCHI, *Anagama Vase*, 2022, Porcelain, 24cm x 10cm x 10cm

technology.[5] Their innovative approach involved translating speech volume into a dotted texture on a cylindrical cup.

Ongoing Project

I became particularly interested in finding Anagama data related to my own work as a ceramic artist. Since I joined a wood-firing group in 2022, I've noticed various changes and events that occur during the continuous five-day stoking period. In addition, every firing is different. The process often takes place under changing weather conditions and involves different crew members in multiple phases. I thought it would be fascinating to build a vessel based on those temperature changes, with a form resembling a flame tornado.

We captured temperature data for nine days, including the cool-down period. Observing the fluctuating temperature trends, I envisioned creating a vase by generating a spiral shape based on the time sequence. Since the device recorded data every minute, I used 60 data points for each layer of the 3–D printed vase. In other words, each layer represents an hour's worth of activity. The final design of the vase comprises around 204 layers. Although the pre-print preview showed a printable shape, the initial version collapsed during printing. This failure occurred because the wireless transmitter frequently lost connection during the firing, leading to glitches in the layers. To address this, I returned to the original spreadsheet and filled in any missing values – indicated by zeros – using simple linear interpolation. After this smoothing process, the 3–D printer operated without issues.

I'm excited to bring this piece to another Anagama kiln or to start printing firing data from other sites. This project holds a metaphysical meaning for me: it involves capturing data from one Anagama and then firing the resulting artwork in a different Anagama.

Methodology

When examining the steps involved in 3–D printing, the slicing process is the last step before sending the data to the printer. Here, the 3–D model is sliced into multiple layers, enabling the printer head to move along the contour outlines. This output format, known as G-code (RS-274), is simple to understand in plain text format, using X–Y coordinates and a Z coordinate for height. The instructions are listed line-by-line and are self-explanatory. This can be edited in spreadsheet programs like Microsoft Excel, or I could develop a user-friendly web application, which has been published as an open-source project.[6]

In extending this principle to more advanced forms, such as my project on global warming data, the center coordinate serves as a reference point around which rotated points are calculated. In my example, the data was collected on a monthly basis, so I divided 360°s by 12 to represent the temperature for each month. Each segment then corresponds to a 30° angle. The calculations for the next coordinates are as follows, where n is the number for the month:

X coordinate = offset X + distance * cos (30° * n)
Y coordinate = offset Y + distance * sin (30° * n)

In this figure, the black segments represent the shape if the value is consistent, maintaining the same length at each corner. Meanwhile, the red segments illustrate how the outer contour line adjusts according to the data values.

Closing Comment

Drawing from my dual background in art and computer science, I've discovered that 3–D clay printers offer more than just standard, cookie-cutter outputs. Utilizing my expertise in data visualization, I've crafted ceramics where "the data dictates the form," thereby making the technology more accessible and lowering technical barriers. My works have been showcased in mixed-media exhibitions in a Manhattan gallery, a presentation at the New England Wood Firing Conference, and an article by the Museum of Modern Ceramic Art in New York. The feedback I've received has been constructive, sparking new ideas for future projects. My ongoing goal is to further explore this unique approach to ceramic art, expanding beyond time-series data to incorporate other forms of data, such as frequency distribution or topographic information.

Kiichi Takeuchi has built more than 165 apps, and he has been in the computer science field for more than 20 years. He has been studying ceramic art as a BFA at Long Island University. In the studio, he is in charge of 3–D printers for research, helping students, and maintenance.

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